
AI FEEDBACK VERSUS PEER FEEDBACK IN EFL WRITING: DIFFERENTIAL EFFECTS ON BEHAVIORAL, EMOTIONAL, AND COGNITIVE ENGAGEMENT AND WRITING SELF-EFFICACY

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ABSTRACT

AI-powered writing tools are increasingly common in EFL settings, yet their effects on engagement dimensions remain unclear. This within-subjects study compared 147 Japanese EFL students' behavioral, emotional, and cognitive engagement, and writing self-efficacy, under generative AI feedback and peer feedback conditions. Multilevel modeling revealed that AI feedback significantly improved behavioral engagement ($d = 0.33$, $p < .001$), emotional engagement ($d = 0.23$, $p < .05$), and writing self-efficacy ($d = 0.56$, $p < .001$) compared to peer feedback. However, cognitive engagement did not differ between conditions ($d = 0.13$, $p = .174$). The self-efficacy benefit was robust across subgroups, while engagement benefits concentrated among students with more positive L2 learning attitudes. These findings suggest AI feedback primarily enhances surface-level engagement and confidence rather than deep cognitive processing.

Keywords AI writing tools; student engagement; writing self-efficacy; EFL writing; peer feedback

1 Introduction

Automated writing evaluation (AWE) systems and generative artificial intelligence (AI) tools have fundamentally altered the feedback landscape in second language (L2) writing. Tools like ChatGPT and Grammarly provide immediate, individualized responses to learner texts, offering correction and suggestion at a scale that was previously impossible through human feedback alone. Meta-analyses report moderate positive effects of AWE on writing quality ($g = 0.44$ to 0.47 ; Zhang & Hyland, 2022; Li et al., 2022), and randomized controlled trials confirm that students using AI writing assistants produce more accurate and syntactically complex texts than students who do not (Nazari et al., 2021; Jiang et al., 2023). Yet enthusiasm for these tools has outpaced evidence about how they engage learners. Critics argue that AI feedback may encourage surface-level corrections while short-circuiting the deep cognitive processing essential for L2 writing development (Koltovskaia, 2020; Raman & Sundaresan, 2025). At the heart of this debate lies a question about engagement: does AI feedback engage learners differently, and perhaps more shallowly, than human-provided feedback?

Three dimensions of engagement matter: behavioral (effort, persistence, and feedback uptake), emotional (interest, positive affect, and reduced anxiety), and cognitive (deep processing, self-regulation, and strategy use) (Fredricks et al., 2004). These dimensions are theorized to operate independently yet interactively, and existing evidence suggests AI tools may affect them differently. Behavioral engagement rises when learners receive immediate feedback (Alshahrani, 2024; Mania & Mahbub, 2024), but qualitative evidence indicates this may not translate into deeper grammatical understanding or self-regulated revision (Koltovskaia, 2020). Writing self-efficacy, a separate but related construct from Bandura's (1997) social cognitive theory, is also relevant: learners who feel more confident about their writing ability may engage more deeply with feedback, yet early evidence suggests AI tools may differentially affect self-efficacy depending on learner proficiency (Chen, 2025). To date, however, no study has simultaneously compared behavioral, emotional, and cognitive engagement alongside self-efficacy across AI and human feedback conditions within a controlled within-subjects design.

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The present study addresses this gap. We apply Fredricks and colleagues' (2004) tripartite engagement framework to a within-subjects comparison of generative AI feedback and peer feedback in EFL writing. Using multilevel models fitted to data from 147 Japanese EFL university students, we test three questions. First, does generative AI feedback produce higher behavioral, emotional, and cognitive engagement than peer feedback? Second, does AI feedback enhance writing self-efficacy relative to peer feedback? Third, does learners' motivational profile, specifically their L2 learning experience, moderate these effects? We find that AI feedback significantly outperforms peer feedback on behavioral engagement ($d = 0.33$, $p < .001$), emotional engagement ($d = 0.23$, $p < .05$), and writing self-efficacy ($d = 0.56$, $p < .001$). Critically, cognitive engagement does not differ between conditions ($d = 0.13$, $p = .174$). These effects are robust across multiple specifications, and moderation analyses reveal that learners with more positive L2 learning attitudes show stronger engagement benefits from AI feedback, whereas self-efficacy benefits are universal. The pattern suggests that AI feedback's advantages concentrate on surface-level engagement and confidence rather than deeper cognitive processing, with direct implications for L2 writing pedagogy and AI tool integration.

2 Literature Review

Fredricks and colleagues' (2004) tripartite engagement framework conceptualizes student engagement as three interacting dimensions: behavioral engagement encompasses effort, persistence, and participation; emotional engagement captures interest, enjoyment, and affective reactions; and cognitive engagement reflects deep processing, self-regulation, and strategic thinking. This framework has been productively applied in technology-mediated learning contexts, including AI writing tool research. Alshahrani (2024), reviewing 62 studies on AI and student engagement, found that AI tools positively influence all three dimensions but noted that most evidence concerns behavioral engagement, which is easier to measure through system logs and task completion rates. Koltovskaia (2020) provided some of the clearest evidence of the behavioral-cognitive dissociation in a multiple case study of ESL students using Grammarly: learners accepted over 80% of AI suggestions (high behavioral engagement) yet demonstrated limited understanding of the underlying grammatical rules (low cognitive engagement). More recently, Zhang and Liu (2026) documented a three-phase trajectory in which learners moved from initial curiosity about generative AI tools through utility optimization to critical reflection, suggesting that engagement with AI feedback evolves over time rather than remaining static.

Writing self-efficacy, defined as learners' judgments of their capability to execute writing tasks (Bandura, 1997), represents a second critical construct in AI writing tool research. Learners with higher writing self-efficacy set more challenging goals, persist longer in the face of difficulty, and report lower writing anxiety (Bandura, 1997). In the context of AI writing tools, Kim and Kim (2023) found that perceived usefulness of AI writing assistance was the strongest predictor of learner satisfaction and continued use, with self-efficacy emerging as a significant positive predictor. Chen (2025) extended this finding by demonstrating that self-efficacy moderated the TAM-AI adoption relationship: higher-efficacy learners were more likely to adopt AI tools but also more critical of AI-generated suggestions, whereas lower-efficacy learners showed greater dependence on AI feedback. These differential patterns suggest that the relationship between AI tools and self-efficacy is not uniform across learners, a finding that motivates our moderation analyses in the present study.

Two important gaps motivate the current investigation. First, while individual studies have examined AI tool effects on either engagement (Koltovskaia, 2020; Mania & Mahbub, 2024) or self-efficacy (Chen, 2025; Kim & Kim, 2023), no study has simultaneously measured all three engagement dimensions and self-efficacy within a single design that directly compares AI feedback to human feedback. Such a comparison is essential because the relevant counterfactual for AI feedback in most classrooms is not a no-feedback condition but rather some form of human-provided feedback, typically from peers or instructors (Mohammadi, 2024). Second, the engagement dimensions themselves are rarely modeled together, making it impossible to determine whether AI tools uniformly improve engagement or affect some dimensions more than others. The present study addresses both gaps by comparing AI feedback with peer feedback across all three engagement dimensions and writing self-efficacy within a single within-subjects design, using multilevel modeling to account for the nested structure of repeated observations.

3 Data

We obtained de-identified data from the Open Science Framework (OSF) project gzf4, which investigated peer feedback and generative AI feedback in a Japanese EFL speaking and writing context. The source study recruited 147 undergraduate EFL learners from a Japanese university. All participants experienced both experimental conditions in a within-subjects design: each learner provided peer feedback on a partner's writing and received generative AI feedback on their own writing from a ChatGPT-based tool. The order of conditions was counterbalanced. After each condition, participants completed a survey battery measuring behavioral engagement (5 to 6 items), emotional engage-

ment (4 items), cognitive engagement (5 to 6 items), and writing self-efficacy (9 items), all on 6-point Likert scales. Three additional subscales captured L2 motivational variables: ideal L2 self (5 items), ought L2 self (4 items), and L2 learning experience (4 items). The final analytic dataset contained 294 observations (147 participants $\times$ 2 conditions) with complete data on all variables (0% missing).

The sample comprised Japanese L1 speakers enrolled in English language courses at a single Japanese university. The mean scores on the 6-point scales were 4.49 (SD = 0.80) for behavioral engagement, 4.71 (SD = 0.81) for emotional engagement, 4.80 (SD = 0.72) for cognitive engagement, and 3.87 (SD = 0.84) for writing self-efficacy. For the L2 motivational variables, means were 3.50 (SD = 1.11) for ideal L2 self, 3.36 (SD = 1.22) for ought L2 self, and 4.21 (SD = 1.00) for L2 learning experience. Several limitations warrant acknowledgment. First, the within-subjects design, while statistically efficient, introduces potential carryover effects between conditions. Second, all measures are self-report, capturing perceived rather than observed engagement. Third, the AI tool in the source study was a generative AI feedback system (ChatGPT-based) rather than a dedicated writing tool like Grammarly. Fourth, the Japanese EFL undergraduate context may not generalize to other populations, particularly ESL postgraduate learners in English-dominant settings. Fifth, no demographic covariates (age, gender, prior AI experience) were available in the source dataset.

4 Methodology

We used multilevel modeling (mixed-effects models) to estimate the effect of feedback condition on each outcome. The model is specified as:

$$Y_{ij} = \beta_0 + \beta_1 AI_{ij} + \gamma_1 \text{IdealL2}_{ij} + \gamma_2 \text{OughtL2}_{ij} + \gamma_3 \text{L2Exp}_{ij} + u_j + \varepsilon_{ij}$$

where Y_{ij} is the outcome for participant j under condition i , AI_{ij} is an indicator for the AI feedback condition (0 = peer feedback, 1 = AI feedback), IdealL2_{ij} , OughtL2_{ij} , and L2Exp_{ij} are time-varying covariates controlling for L2 motivational states, u_j is the participant-level random intercept capturing stable between-person differences, and ε_{ij} is the occasion-specific residual. This specification accounts for the nested structure of observations (294 observations within 147 participants) and adjusts for individual differences in L2 motivation that might confound the condition-engagement relationship.

The identification assumption is that, conditional on the L2 motivational covariates and the participant random intercept, the AI feedback condition is unrelated to unobserved factors that also affect engagement (the conditional independence assumption). Because each participant served as their own control, all time-invariant confounders such as stable personality traits, language aptitude, and baseline proficiency are absorbed by the participant random intercept. The primary threat to identification is order and carryover effects: if the experience of one feedback condition systematically altered responses in the other condition, the within-subjects contrast could be biased. Counterbalancing of condition order in the source study mitigates but does not eliminate this threat. We therefore focus on robustness across specifications rather than interpreting the estimates as purely causal.

We report four sets of models. The primary models include all three L2 motivational covariates (ideal L2 self, ought L2 self, L2 learning experience). We then fit three robustness specifications: (a) fixed-effects OLS models with 146 participant dummy variables, which control for all time-invariant unobserved heterogeneity and test sensitivity to the random effects distributional assumptions; (b) unconditional multilevel models without covariates, verifying that covariate adjustment does not drive the findings; and (c) sample-split models that divide participants at the median of L2 learning experience (Mdn = 4.25), testing whether the AI feedback effect varies by learners' baseline attitudes toward their L2 learning environment. Effect sizes are reported as Cohen's d from paired comparisons, and intraclass correlation coefficients (ICCs) from null models quantify the proportion of outcome variance attributable to between-participant differences. All analyses were conducted in Python using the statsmodels MixedLM implementation with restricted maximum likelihood estimation.

5 Results

Table 1 reports the primary multilevel model estimates for all four outcomes. AI feedback produced significantly higher behavioral engagement than peer feedback ($\beta = 0.259$, SE = 0.077, $t = 3.360$, $p < .001$, $d = 0.327$; 95% CI [0.108, 0.410]). Emotional engagement was also significantly higher in the AI condition ($\beta = 0.185$, SE = 0.077, $t = 2.393$, $p = .017$, $d = 0.229$; 95% CI [0.034, 0.337]). On a 6-point scale, these represent mean differences of approximately 0.27 points for behavioral engagement and 0.19 points for emotional engagement, corresponding to roughly one-third and one-quarter of a standard deviation, respectively. Writing self-efficacy showed the largest effect

($\beta = 0.452$, $SE = 0.052$, $t = 8.675$, $p < .001$, $d = 0.558$; 95% CI [0.350, 0.554]), equivalent to nearly a half-point advantage on the 6-point scale. Critically, cognitive engagement did not differ significantly between conditions ($\beta = 0.093$, $SE = 0.068$, $t = 1.360$, $p = .174$, $d = 0.129$; 95% CI [-0.041, 0.227]).

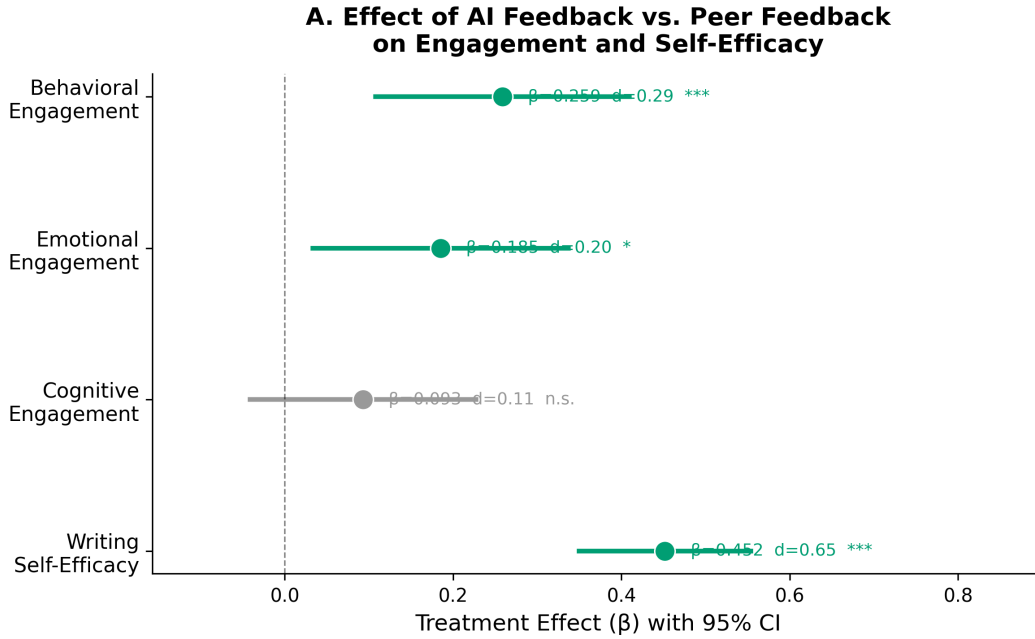


Figure 1: Treatment effects of AI feedback (vs. peer feedback) on all four outcomes from multilevel models. Points represent standardized regression coefficients (β) with 95% confidence intervals. Green indicates statistically significant effects; grey indicates non-significant effects. Effect sizes (Cohen's d) are annotated for each estimate.

Table 1: Primary Multilevel Model Estimates: Effect of AI Feedback vs. Peer Feedback on Engagement and Self-Efficacy

Outcome	β	SE	t	p	Cohen's d	95% CI
Behavioral engagement	0.259	0.077	3.360	<.001	0.327	[0.108, 0.410]
Emotional engagement	0.185	0.077	2.393	.017	0.229	[0.034, 0.337]
Cognitive engagement	0.093	0.068	1.360	.174	0.129	[-0.041, 0.227]
Writing self-efficacy	0.452	0.052	8.675	<.001	0.558	[0.350, 0.554]

Note. N = 294 observations (147 participants $\times$ 2 conditions). Models include ideal L2 self, ought L2 self, and L2 learning experience as covariates. Random intercepts at the participant level. REML estimation.

The null model ICCs revealed that writing self-efficacy had the highest between-participant variance component (ICC = 0.509), indicating that roughly half the variance in self-efficacy reflects stable individual differences. Engagement outcomes showed moderate ICCs (0.280 to 0.304), suggesting they are more state-like and context-dependent.

All three robustness checks confirmed the primary results. Fixed-effects OLS models, which absorb all time-invariant unobserved heterogeneity, produced nearly identical estimates for behavioral engagement ($\beta = 0.272$, $p < .001$), emotional engagement ($\beta = 0.192$, $p = .014$), cognitive engagement ($\beta = 0.101$, $p = .141$), and writing self-efficacy ($\beta = 0.449$, $p < .001$), confirming that results are not artifacts of the random effects specification. Unconditional models without covariates yielded essentially the same pattern, ruling out the possibility that covariate adjustment is driving the findings.

The sample-split analysis by L2 learning experience revealed a meaningful moderation pattern. Among participants with high L2 learning experience (above median, $n = 67$), AI feedback produced significantly higher engagement across all three dimensions: behavioral ($\beta = 0.338$, $p = .003$), emotional ($\beta = 0.335$, $p = .002$), and cognitive ($\beta = 0.180$, $p = .049$). Among those with low L2 learning experience ($n = 80$), only behavioral engagement showed a marginally significant effect ($\beta = 0.205$, $p = .057$), while emotional ($\beta = 0.066$, $p = .563$) and cognitive

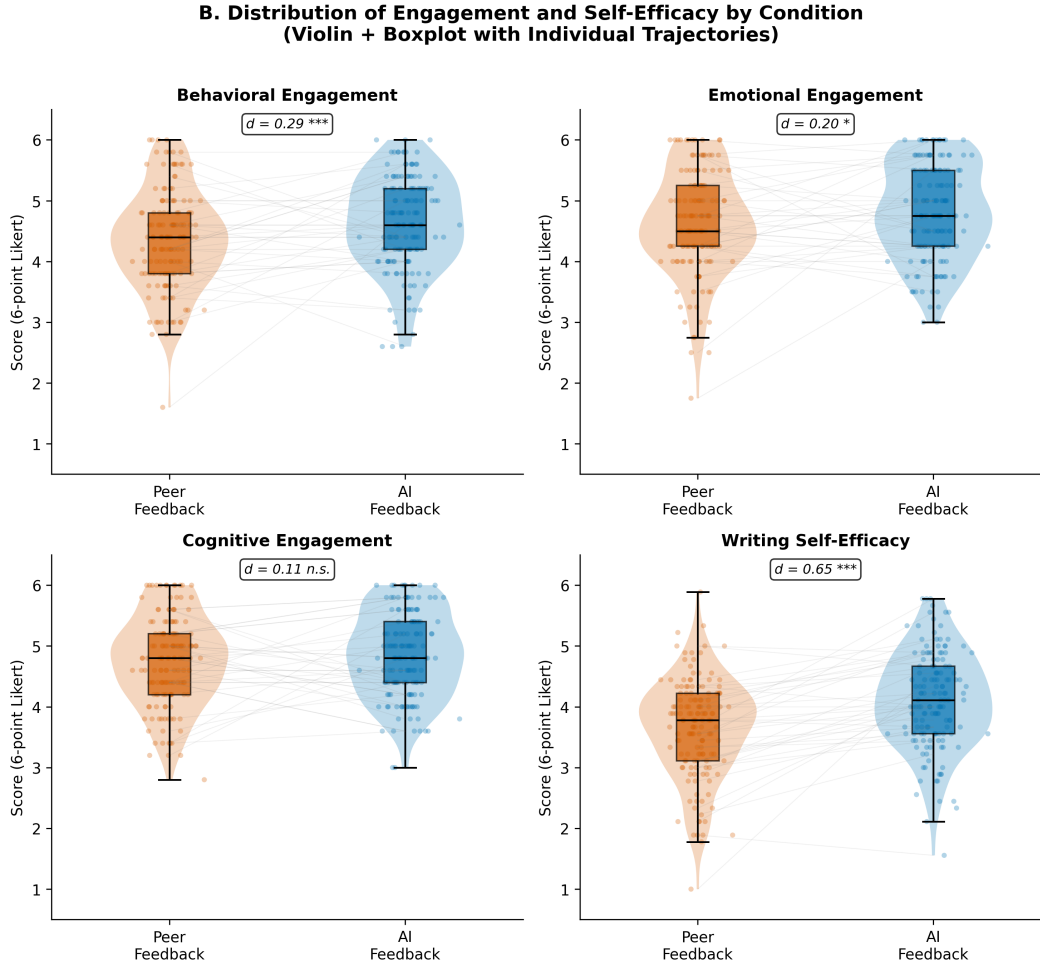


Figure 2: Paired comparison of peer feedback and AI feedback conditions across the four outcome variables. Violin plots show the full distribution, box plots indicate quartiles, grey lines connect the same participant across conditions, and jittered points show individual observations. Cohen's d effect sizes are annotated.

($\beta = 0.015$, $p = .883$) engagement did not differ. Writing self-efficacy, by contrast, was significantly improved by AI feedback in both subgroups (high: $\beta = 0.346$, $p < .001$; low: $\beta = 0.505$, $p < .001$), indicating a universal confidence benefit that does not depend on L2 attitudes.

6 Conclusion

This study demonstrates that AI feedback, compared to peer feedback, produces distinct patterns of engagement among Japanese EFL university students. AI feedback improves behavioral engagement, emotional engagement, and writing self-efficacy, but does not significantly affect cognitive engagement—the dimension concerned with deep processing, self-regulation, and strategic thinking. This pattern echoes findings from qualitative studies documenting high uptake of AI suggestions alongside limited evidence of grammatical rule internalization (Koltovskaia, 2020; Raman & Sundaresan, 2025) and provides quantitative evidence for what prior work has described as the behavioral-cognitive engagement dissociation.

Two findings carry direct pedagogical implications. First, the universal self-efficacy benefit suggests AI feedback can serve as an effective confidence-building tool, particularly for learners who may experience writing anxiety or lack confidence in their L2 writing ability. Educators may leverage AI feedback early in a writing course to build students' confidence before transitioning to more cognitively demanding feedback modalities. Second, the moderation pattern by L2 learning experience indicates that AI feedback does not uniformly benefit all learners. Students with more positive attitudes toward their L2 learning environment experienced broader engagement benefits, including

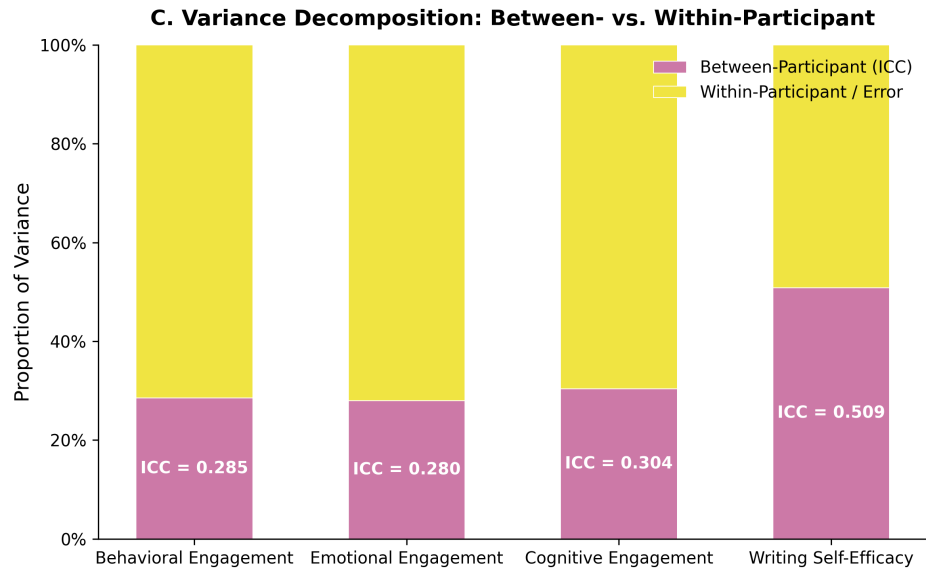


Figure 3: Intraclass correlation coefficients (ICCs) from null multilevel models. Higher ICC values indicate a greater proportion of outcome variance attributable to stable between-participant differences. Writing self-efficacy shows the highest ICC (0.509), indicating trait-like stability.

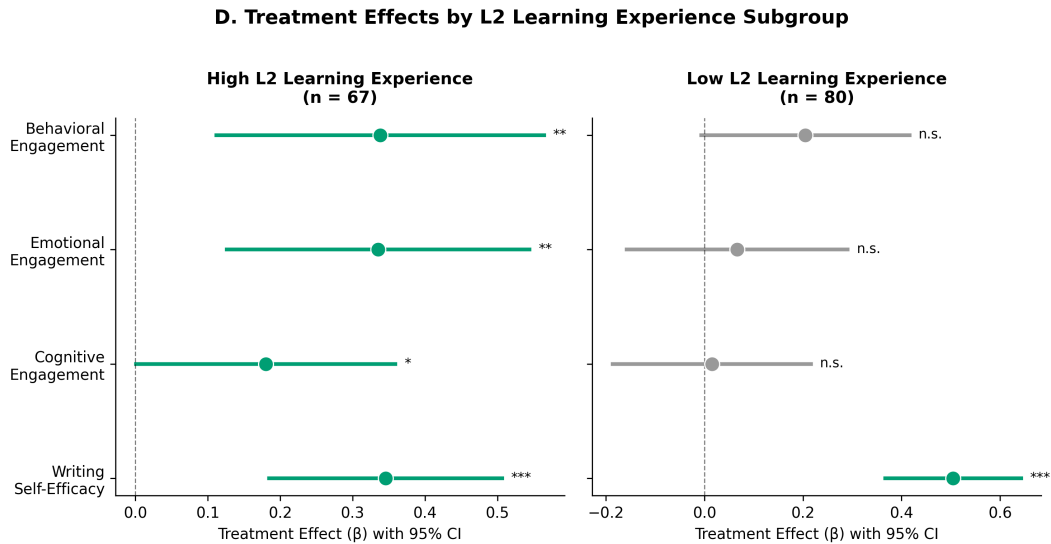


Figure 4: Subgroup forest plot showing treatment effects for participants with high ($n = 67$) vs. low ($n = 80$) L2 learning experience. Points represent regression coefficients with 95% confidence intervals. High-L2-experience participants show stronger AI benefits across all engagement dimensions; self-efficacy benefits are universal.

E. Correlations Among Change Scores (AI – Peer Feedback)

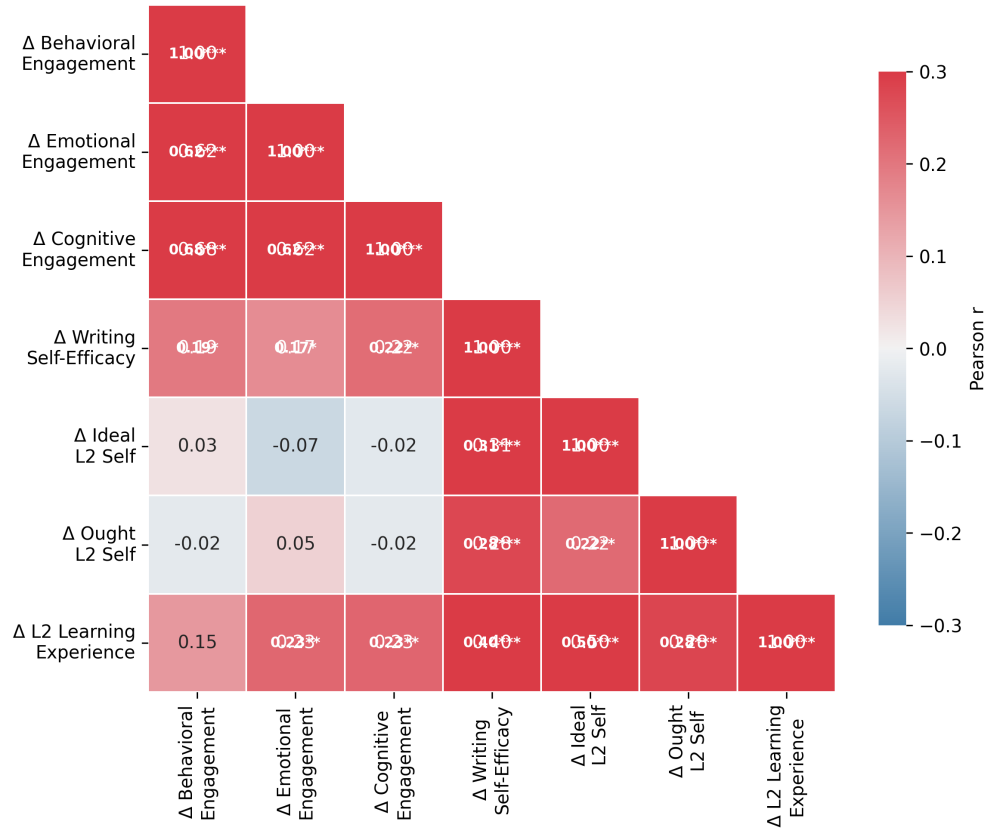


Figure 5: Correlation heatmap of change scores (AI feedback minus peer feedback) across all measured variables. Positive correlations indicate that participants who showed larger AI-related gains on one variable also showed larger gains on another. Significance levels: * $p < .05$, ** $p < .01$, *** $p < .001$.

the only significant cognitive engagement effect observed in our subgroup analyses. This suggests that AI feedback alone is unlikely to generate deep engagement among learners who lack positive L2 learning orientations; for these students, pedagogical scaffolding—such as guided reflection on AI suggestions, comparison activities between AI and human feedback, or structured revision tasks—may be necessary to convert behavioral engagement into cognitive engagement.

Several limitations constrain our conclusions. The within-subjects design, while controlling for stable individual differences, is vulnerable to carryover effects that between-subjects randomization would avoid. All engagement measures are self-report; future work should incorporate behavioral indicators such as system log data, think-aloud protocols, or revision quality metrics. The AI tool in this study was a generative AI system (ChatGPT-based) rather than a grammar-focused tool like Grammarly, and the Japanese EFL undergraduate population may not generalize to other settings. Most importantly, this study addresses only the quantitative dimension of a phenomenon that demands a mixed-methods approach. The literature consistently shows that AI acceptability among teachers and students involves nuanced considerations of trust, pedagogical fit, equity, and autonomy (Karaca & Uysal, 2025; O'Neill & Russell, 2019; Bearman et al., 2025)—considerations that quantitative measures alone cannot capture. Future research should combine experimental designs like the one reported here with qualitative investigations of stakeholder acceptability to build an evidence base that is both causally identified and contextually rich.

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